AI for Good Sandbox Hackathon – إِتْقَان – Micro-Assessment AI Agent

- Team leader: Toqa Mahmoud Tawfiq Al-zoubi
- Team_members: 1. Afnan Bassem Mohammad Sami Sawalha
  2. Natalia Samer Mohammad Abu Durei
- Contact details: alzobitoqa@gmail.com
- Github link: https://github.com/alzoubitoqa/micro-assessment-agent.git

1. Introduction

   Education systems globally rely heavily on high-stakes, spaced-out exams to evaluate student performance. However, this traditional approach fails to provide immediate feedback to students and delays instructors from identifying knowledge gaps until it is too late for early intervention.

   We propose the **Micro-Assessment AI Agent**, an intelligent, proactive formative assessment system. It transforms academic evaluation into a continuous, adaptive, and interactive process. The system utilizes a Multi-Agent Workflow to ingest lecture materials (PDFs), generate personalized micro-quizzes, adapt to student performance in real-time (providing hints and simplified questions), and furnish instructors with a live "Knowledge Gap Heatmap." Furthermore, it includes an Integrity Agent to ensure academic honesty during remote assessments.

   Our solution directly aligns with the ITU's "AI for Good" mission by utilizing AI to enhance the quality of education (SDG 4), reduce student anxiety, and democratize personalized learning at scale.

2. Description of the use case

   The Problem: Instructors currently lack real-time visibility into student comprehension. Furthermore, static, high-stakes exams increase student stress and offer no adaptive feedback to help correct misunderstandings during the learning process. Moreover, remote assessments face significant integrity challenges.

   The Solution: Our solution bridges these gaps by introducing an autonomous AI Agent pipeline:

- **Dynamic Generation (RAG):** Replaces static question banks. The system extracts core concepts from lecture slides and generates context-aware questions on the fly, ensuring strict alignment with the curriculum (Zero-Hallucination via a Validator Agent).
- **Adaptive Feedback Loop (LangGraph):** If a student answers incorrectly, the Agent provides a contextual hint. If they fail again, it simplifies the question. This ensures the student learns from the assessment rather than just being penalized.
- **Forensics & Integrity Agent:** Analyzes response times to detect suspiciously fast answers (indicative of AI copy-pasting), ensuring fair remote evaluation.

- **Instructor Heatmap:** Visualizes class-wide data in real-time, highlighting specific concepts (in red) where the majority of students struggle, allowing the instructor to adjust their next lecture.

Multi-Agent Roles in Our Architecture:

- **Role-based Agents:** The *Question Generator* (acts as an examiner) and the *Validator Agent* (acts as a strict reviewer to ensure zero-hallucination).
- **Coordinating Agent:** The *LangGraph State Machine*. It acts as the orchestrator, deciding whether to route the student to a "Hint", a "Simplified Question", or to "End the session".
- **Policy Agent:** The *Integrity Agent*, which operates on pre-defined academic honesty policies (e.g., flagging answers completed in <3 seconds) and alerts the human instructor (Human-in-the-loop) without taking autonomous punitive action.

3. Use case requirement

- **Data Node (M):** The system requires digital lecture materials (PDFs). Text is chunked and embedded locally using Sentence-Transformers to create a private vector database (ChromaDB), ensuring proprietary educational content is not leaked to public LLMs.
- **Model Node (M):** We utilize Groq API (Llama-3.3-70b) for ultra-fast generation and validation. LangGraph is used to orchestrate the state machine.
- **Data Privacy (P):** Student data (names, answers, response times) is stored locally in a secure SQLite database. No personally identifiable information (PII) is sent to the LLM; only the lecture context is passed for question generation.
- **Edge vs. Cloud:** The system operates on a cloud-based Streamlit architecture, accessible via a web browser, making it highly scalable for universities without requiring heavy edge-computing hardware.

Our ML Pipeline Architecture:

1. **Data Source:** Lecture PDFs uploaded by instructors.
2. **Data Aggregation:** Text extraction and semantic chunking.
3. **Preprocessing:** Local embedding using Sentence-Transformers (Privacy preserved).
4. **Model:** Groq API (Llama-3) for generation and ChromaDB for vector retrieval.
5. **Inference:** LangGraph orchestrates the LLM to generate and validate questions.
6. **Policies (Guardrails):** Validator Agent prevents hallucinations; Integrity Agent enforces time-based honesty.
7. **Distribution & Target:** Data is distributed to the Instructor Heatmap and Student Portal.

4. Mapping with ITU-T Y.3172 Readiness 2.0 Report, and domain-specific policies

Node	Use case (Education Domain)	Readiness 2.0 Report	Documents (e.g., policies, strategies)
SRC	Instructors upload lecture PDFs. Students provide answers via the web interface.	Data and model marketplace	UNESCO Recommendation on the Ethics of AI (2021), Article 4; Institutional copyright guidelines.
C	Web infrastructure (Streamlit) and API calls (Groq).	Edge AI usage	UAE National Strategy for AI 2031 (Digital Infrastructure Pillar); HTTPS & API reliability standards.
PP	Text chunking and local embedding (Sentence-Transformers).	Data harmonization	**Privacy:** Local processing ensures lecture data and student PII remain within the institutional server (GDPR & UAE Data Protection Law).
M	LLM (Llama-3), Concept Extractor, Question Generator, Integrity Agent.	Source of data input, source of model	**Fairness:** Validator Agent prevents biased questions. **Integrity:** Time-based forensic checks.
P	LangGraph decision routing (Hint vs. Simplify). Heatmap generation.	Trustworthiness of the model	Human-in-the-loop guidelines; The instructor reviews Heatmap and Integrity Flags before academic action.
D	ChromaDB (Vector) and SQLite (Relational) hosted on the university server.		Institutional Data Retention Policies; Secure storage standards (ISO 27001).
SINK	Instructor Dashboard (Heatmap) and Student Portal (Quizzes).	Collaboration with AI, human in the loop	**Usability:** Accessible UI (WCAG standards). **Ethics:** AI flags are advisory; instructors make final decisions.

Alignment with AI Readiness 13 Dimensions:

- **Contextualization:** Adaptable to any university curriculum by generating questions directly from uploaded PDFs.
- **Human Interface:** Intuitive Streamlit web interfaces for ease of use.
- **Strategy Alignment:** Supports SDG 4 (Quality Education) and modernizes assessment strategies.
- **Impact of Humans in AI Integration:** The AI acts as an advisory tool; instructors retain full authority to act on Integrity flags.
- **Inclusion:** Adaptive hints and simplified questions ensure students with varying comprehension levels are supported.
- **Digital Infrastructure:** Cloud-based, requiring no heavy edge-computing hardware.

5. Error scenarios

Scenario: Integrity Agent False Positives (Controversy)

- **Step-1:** The system is deployed in a university. A student with a photographic memory or fast reading skills answers a question correctly in 2.5 seconds.
- **Step-2:** The Integrity Agent flags the session as "Suspicious" (due to the <3 second threshold), and the instructor sees a red flag.
- **Step-3:** If the instructor immediately penalizes the student, this causes a grievance.
- **Resolution:** Based on our policy design, the Integrity Agent is a "Human-in-the-loop" advisory tool. The system does not automatically fail the student. Instead, the instructor uses the flag to initiate a conversation or a proctored re-test. This protects academic integrity without violating the student's rights to due process.

Scenario: Hallucination / Irrelevant Question

- **Step-1:** An uploaded PDF has formatting errors.
- **Step-2:** The LLM generates a question that is factually incorrect based on the lecture.
- **Resolution:** Our built-in **Validator Agent** double-checks the generated question against the source context before showing it to the student. If the validation fails, the system regenerates the question or aborts the process.

6. Evaluation Scenario: The "Continuous Feedback" Case

- **Setup:** Dr. Sarah uploads her lecture PDF to the Instructor Dashboard.
- **Action:** Student Ali logs in, starts a quiz, and answers incorrectly.
- **AI Reaction:** The Coordinating Agent (LangGraph) routes him to a contextual Hint. When he fails again, it generates a Simplified Question.
- **Integrity Check:** Ali answers a different question correctly in 1.5 seconds. The Integrity Agent flags this anomaly.

- **Result:** Dr. Sarah sees a Red Heatmap indicating the class struggled with the concept. She reviews Ali's Integrity flag, deciding to discuss it with him (Human-in-the-loop). No automatic punishment is applied.

7. Policy gaps and standards gaps derived from the mapping exercise

- **Data Privacy for Minors:** Current AI policies often lack strict guidelines on how LLMs should handle minors' interaction data in educational tools. (Addressed by our system by keeping PII out of the LLM prompt).
- **AI-Assisted Academic Integrity:** There is a gap in standardized policies regarding how much AI analysis (like time-tracking) is permissible before it infringes on student privacy. Clear guidelines are needed to differentiate between "learning analytics" and "surveillance".
- **Algorithmic Fairness in Testing:** Standards must be developed to ensure AI-generated questions do not carry linguistic or cultural biases, which our Validator Agent currently mitigates internally but requires external auditing standards.